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GEO 266 – GIS Analysis

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Cascadia Subduction Zone Earthquake – Analyzing Neighborhood Exposure in Portland

1) About the Cascadia Subduction Zone

The Cascadia Subduction Zone is a 700-mile-long fault located approximately 70–100 miles offshore along the Pacific coastline, extending from Northern California to British Columbia, Canada. This fault is capable of producing major earthquakes approximately every 300–500 years, including events exceeding magnitude 9.0.

One might ask: if Portland is over 100 miles from the fault, how would it be affected? The most recent major Cascadia earthquake occurred in 1700, and although it was not directly recorded in North America, it was documented in Japan due to the tsunami it generated (State of Oregon, n.d.).

This leads to what is commonly referred to as “The Big One,” a potential future megathrust earthquake along the Cascadia Subduction Zone. Given the estimated recurrence interval, the Pacific Northwest is considered overdue for such an event. Current estimates suggest there is approximately a 37% chance of a magnitude 7.1 or greater earthquake occurring within the next 50 years.

If such an event occurs, it is expected to produce several minutes of intense ground shaking, potentially lasting over five minutes, which could lead to widespread structural damage, particularly in older buildings, as well as significant infrastructure failure.

2) Objective and Research Question

This project is guided by the following research questions:

Which neighborhoods are at the highest risk of impact from a Cascadia Subduction Zone earthquake (“The Big One”), and which populations have the greatest exposure?

To address this objective, four key factors were analyzed:

1. Peak Ground Acceleration (PGA)
2. Liquefaction Probability
3. Landslide Probability
4. Population Density

Peak Ground Acceleration (PGA) represents the **maximum intensity of ground shaking** experienced at a given location during an earthquake. Liquefaction occurs when **loose, water- saturated soils temporarily lose strength and behave like a liquid** due to increased pore water pressure from seismic shaking. Landslides occur when **slopes become unstable**, causing soil, rock, and debris to move downhill.

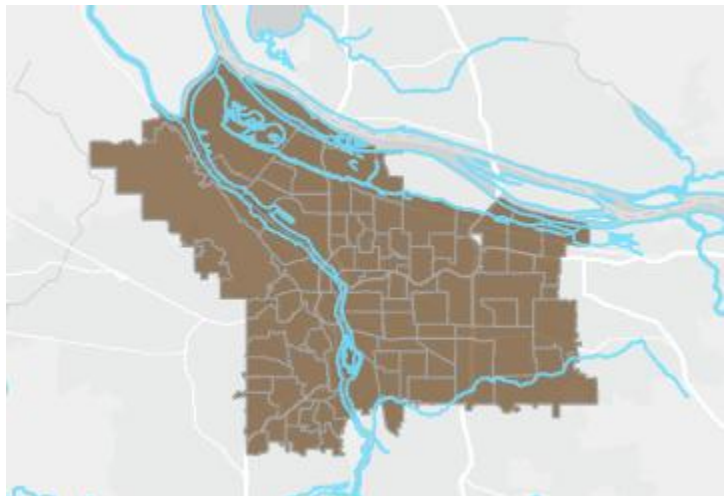
By integrating these three hazard factors with population density, this analysis estimates **neighborhood-level exposure**, identifying areas where earthquake-related hazards are most likely to impact the greatest number of people.

3) Data collection:

All data was collected from three sources – Oregon GEOHub, Portland Open Data and Portland State University.

Dataset/layer name	Data source	Data model (vector, raster)
Peak Ground Acceleration	Oregon GEOHub/DOGAMI	Raster
Liquefaction Probability	Oregon GEOHub/DOGAMI	Raster
Landslide Probability	Oregon GEOHub/DOGAMI	Raster
Neighborhood Boundaries	Portland Open Data	Vector
Population CSV	Civic Life/Portland State University	Vector

Study Area: Portland Neighborhoods



4) Methods used for analysis:

Dataset/layer name	GIS Tools applied to dataset
Project	Neighborhood Shapefile
Extract by Mask	Clips raster layers to the neighborhood shapefile
Zonal Statistics as Table	Used for all three raster layers
Join Field	Combines the raster and neighborhood shapefiles
Reclassify	Converts the raster values to 5 classes
Add field/Calculate Field	Creates hazard/risk scores attributes
Calculate Field	Calculates population to population density
SELECT by Attribute	Used to find the top ten neighborhoods

Project:

The neighborhood shapefile was reprojected from WGS 1984 Web Mercator to NAD 1983 (2011) Oregon Statewide Lambert to ensure consistency with the raster datasets.

Extract by Mask:

The raster datasets covered the entire state of Oregon. The Extract by Mask tool was used to clip each raster to the neighborhood boundary, restricting the analysis to the study area.

Zonal Statistics as Table:

This tool was used to calculate the mean values of each raster within neighborhood boundaries. It was also applied to the reclassified rasters to summarize hazard levels at the neighborhood scale.

Join Field:

The output tables from Zonal Statistics were standalone tables and were joined back to the neighborhood polygon layer to associate hazard values with each neighborhood.

Reclassify:

Raster values were reclassified into five categories: Very Low, Low, Moderate, High, and Very High. This standardization allowed for consistent comparison across hazard types.

Add Field / Calculate Field:

New fields were created to calculate hazard score, risk score, and population density.

- Population Density:
Calculated as population divided by area in square miles:
 $Population\ Density = Population / Area\ (sq\ mi)$
- Hazard Score (CSZ_Score):
Calculated by summing the three reclassified hazard layers:
 $PGA + Liquefaction + Landslide$
- Risk Score (CSZ_Risk):
Calculated by multiplying hazard score by population density:
 $Risk = Hazard \times Exposure$

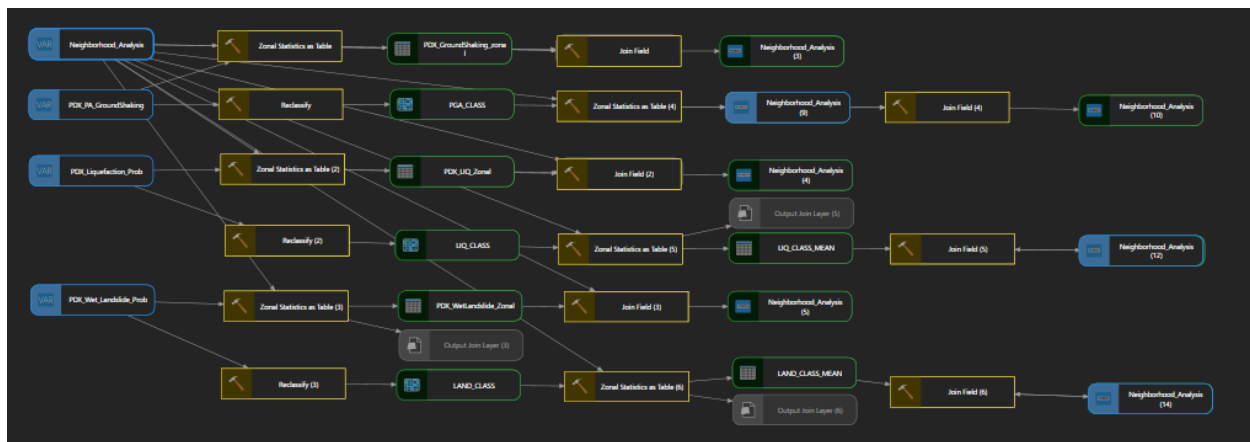
The hazard score represents environmental risk independent of population, while the risk score incorporates exposure by accounting for population density.

Select by Attribute:

The Select by Attribute tool was used to identify the ten neighborhoods with the highest risk scores. These features were then exported for mapping and further analysis.

ModelBuilder:

Most of the analytical workflow was executed within ModelBuilder, allowing for a structured and repeatable process. Exceptions included the Project, Extract by Mask, and Select by Attribute steps, which were performed outside the model.



5) Results:

This project produced seven feature layers:

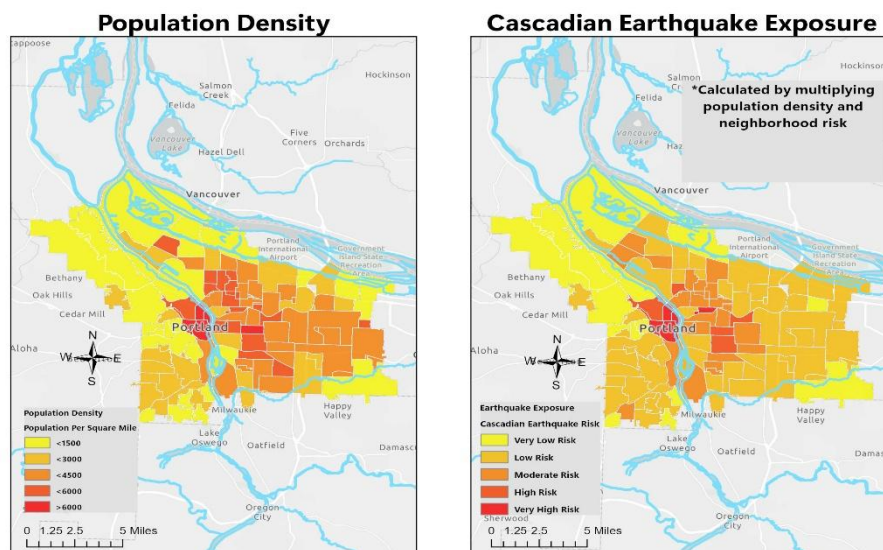
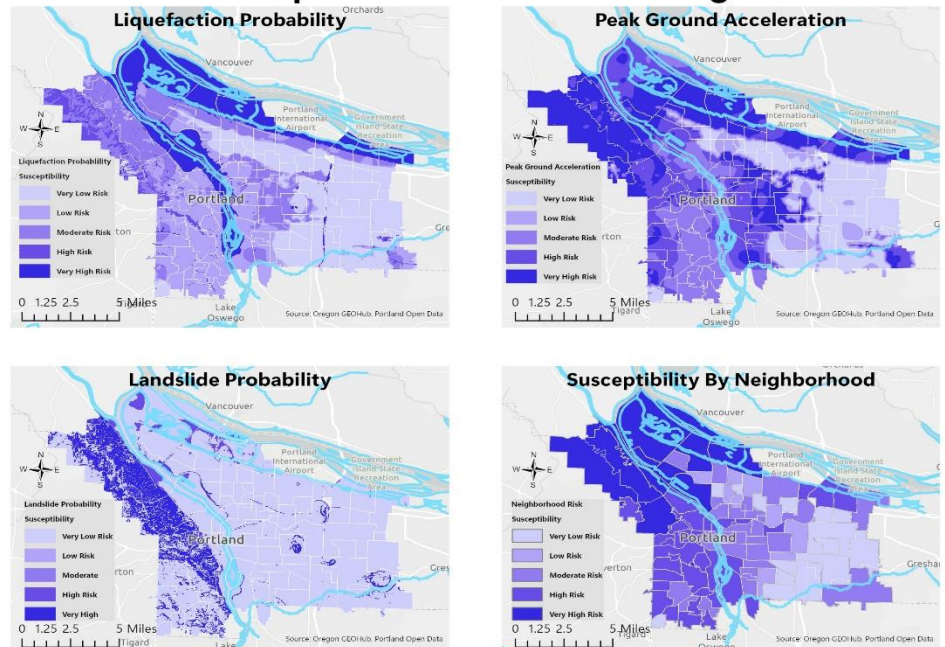
1. Liquefaction Probability
2. Peak Ground Acceleration Probability
3. Landslide Probability
4. Susceptibility by Neighborhood (Hazard Score)
5. Population Density
6. Cascadian Earthquake Exposure (Risk Score)
7. 10 Neighborhoods with the Highest Exposure

Maps 1-4: These maps primarily display the raster-based hazard layers, with the **hazard score** used to symbolize relative levels of hazard at the neighborhood scale.

Overall, western Portland exhibits higher hazard levels compared to the eastern portion of the city. This pattern is largely influenced by steeper topography and more variable elevation, which increases susceptibility to hazards such as landslides. In contrast, areas east of the Willamette River are generally flatter, resulting in lower landslide susceptibility and lower hazard levels.

Additionally, variations in underlying soil conditions contribute to differences in hazard. Areas with looser soil may be more susceptible to effects such as liquefaction during strong ground shaking.

Cascadia Earthquake Risk for Portland Neighborhoods



Maps 5-6: These maps focus on population and exposure. The map on the left displays population density across Portland, while the map on the right combines population density with neighborhood-level hazard to represent overall **exposure**.

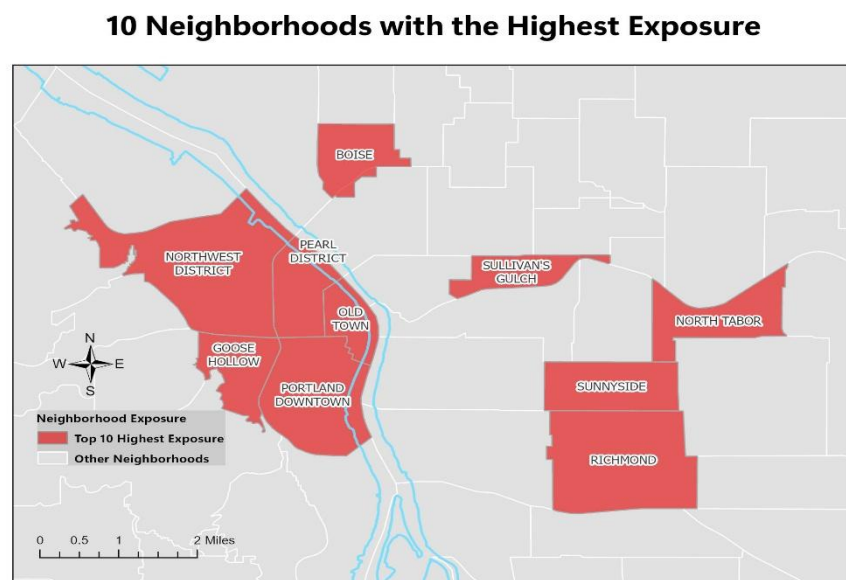
This distinction is important: the exposure map does not simply highlight the most hazardous areas but rather identifies the neighborhoods where earthquake impacts would likely affect the **greatest number of people**. As a result, some areas with moderate hazard but high population density may exhibit higher overall risk than less populated areas with greater environmental hazard.

Final Map: This map highlights the **ten neighborhoods with the highest overall risk** in the event of a Cascadia Subduction Zone megathrust earthquake.

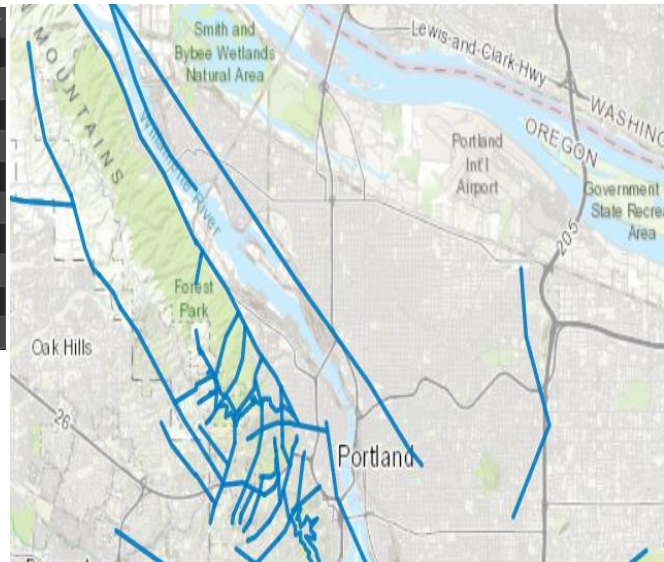
Among these, the **Pearl District** stands out as the most at risk. This is largely due to its high population density combined with a relatively elevated hazard score, resulting in greater overall exposure.

It is also notable that several east side neighborhoods appear among the top ten. While these areas generally exhibit lower landslide susceptibility due to flatter terrain, their inclusion is likely influenced by **higher population densities** and localized geologic conditions.

One possible contributing factor is the presence of the **East Bank Fault**, which runs through portions of the east side. While a Cascadia Subduction Zone earthquake would be the primary driver of regional shaking, such an event could potentially trigger movement on local faults, increasing localized impacts in these areas.



	NAME	CSZ_SCORE	POP_DENS_MILE	CSZ_RISK
1	PEARL DISTRICT	11	12139.829134	133538.120479
2	GOOSE HOLLOW	11	8294.271855	91236.990404
3	OLD TOWN	11	7273.054235	80003.596582
4	PORTLAND DOWNTOWN	10	7358.039466	73580.394657
5	SULLIVAN'S GULCH	12	5982.287532	71787.450388
6	SUNNYSIDE	10	6780.109772	67801.097719
7	NORTHWEST DISTRICT	11	5768.405633	63452.461963
8	BOISE	10	5341.293438	53412.934384
9	RICHMOND	10	5191.32601	51913.260099
10	NORTH TABOR	11	4617.37673	50791.144031



Source: (*Fault Lines*, n.d.)

6) Conclusion and Takeaways

After reviewing the results of this analysis, the research question was effectively addressed. Much of the **downtown core of Portland** exhibits the highest levels of exposure, primarily due to the combination of **high population density and elevated hazard scores**.

As noted earlier, it was somewhat unexpected to see several **east side neighborhoods** rank among the highest in exposure. However, this can likely be explained by the combination of higher population densities and localized geologic factors, including the presence of the **East Bank Fault**.

One additional factor that would be valuable to explore is **hazard intensity independent of population**. While identifying the ten neighborhoods with the highest exposure provides insight into where the greatest number of people may be affected, it does not necessarily indicate where the **most severe physical damage** would occur. For example, several neighborhoods exhibited hazard scores (CSZ_Score) of **13–14**, whereas none of the top ten highest-exposure neighborhoods exceeded a score of **12**. This highlights the distinction between **hazard** and **risk** and suggests that the most hazardous areas are not always the most populated.

If this project were to be expanded, several additional analyses could provide further insight. Mapping the **ten neighborhoods with the highest hazard scores** would help identify areas most susceptible to severe physical impacts. Additionally, analyzing **bridge vulnerability** would be valuable, as Portland contains multiple bridges, some of which may not meet modern seismic safety standards. Finally, examining **building-level vulnerability** could further refine risk estimates, particularly given that over 1,600 structures in Portland may lack the structural integrity to withstand a major Cascadia earthquake (*Portland's Building Structures*, 2023).

Overall, this analysis was both insightful and rewarding, as it allowed me to apply the geospatial tools and techniques learned throughout GEO266 to a real-world hazard scenario. One of the biggest takeaways for me is that **analysis is at the core of GIS**—it not only helps answer specific research questions but also opens the door to new questions and deeper understanding. Working through this project showed me how powerful GIS can be in identifying patterns and relationships that are not immediately obvious. I'm excited to continue developing these analytical skills and apply them to future projects, especially in areas related to hazard assessment and spatial problem-solving.

8) References:

State of Oregon. (n.d.). Cascadia Subduction Zone: Oregon Department of Emergency Management. <https://www.oregon.gov/oem/hazardsprep/pages/cascadia-subduction-zone.aspx>

Neighborhood boundaries. (n.d.). https://gis-pdx.opendata.arcgis.com/datasets/1e95d9b9076742ed9a71de0535ac255c_3/explore?location=45.542750%2C-122.667150%2C11

City of Portland, Oregon 2025. (n.d.). <https://gis-pdx.opendata.arcgis.com/>

Civic Life | PRC Projects. (n.d.). <https://www.prcprojects.us/civic>

Fault lines. (n.d.). <https://geohub.oregon.gov/datasets/oregon-geo::fault-lines/explore?location=45.541473%2C-122.580329%2C11>

Portland's Building Structures. (2023, February).

KOIN. <https://www.koin.com/news/environment/earthquakes/hundreds-of-portland-buildings-have-very-weak-earthquake-defense/>